

Loading required package: `usethis`

```
> devtools::install_github("hadley/lineprof")
Downloading GitHub repo hadley/lineprof@HEAD
...
** testing if installed package can be loaded from temporary
location
** checking absolute paths in shared objects and dynamic
libraries
** testing if installed package can be loaded from final
location
** testing if installed package keeps a record of temporary
installation path
* DONE (lineprof)
```

Consider a function ‘f’ that creates two matrices, doubles its contents, and computes their cross-product. The usage of the ‘lineprof’ function is shown below:

```
> f <- function() {
+   m = matrix(rnorm(30*15000, mean=0, sd=4), 30, 15000)
+   n = matrix(rnorm(30*15000, mean=0, sd=4), 30, 15000)
+   a <- m * 2
+   b <- n * 2
+   crossprod(a, b)
+ }

> l <- lineprof(f())

> l
      time  alloc release dups          ref
src
1 0.005   2.151   0.000   0 c("matrix", "rnorm") matrix/
rnorm
2 0.410 214.577   0.878   0      "crossprod" crossprod
```

We again observe that the ‘crossprod()’ function takes the maximum memory allocation and computation time.

gc()

The `gc()` function triggers the garbage collection to free memory. It also displays information on available memory, as shown below:

```
> gc()
      used (Mb) gc trigger   (Mb) max used   (Mb)
Ncells 294146 15.8   661162  35.4   464793  24.9
Vcells 2457665 18.8  220002509 1678.5 227627845 1736.7
```

tracemem()

The `tracemem()` function is used to debug memory usage issues in R. It can print a message whenever a marked object

for tracing is copied. In the following example, we create an ‘a’ matrix and track its memory usage with the ‘tracemem’ function. The matrix is copied to a ‘b’ matrix, and its contents are then modified. This triggers a printing of the memory address in the output. You can finally stop tracing an R object with the `untracemem()` function, as shown below:

```
> a <- matrix(1:8, nrow=2)
> a
      [,1] [,2] [,3] [,4]
[1,]    1    3    5    7
[2,]    2    4    6    8

> tracemem(a)
[1] "<0x563f0483c808>"

> b <- a

> b[1] <- 10
tracemem[0x563f0483c808 -> 0x563f04838078]:
tracemem[0x563f04838078 -> 0x563f04c850e8]:

> b
      [,1] [,2] [,3] [,4]
[1,]   10    3    5    7
[2,]    2    4    6    8

> a
      [,1] [,2] [,3] [,4]
[1,]    1    3    5    7
[2,]    2    4    6    8

> untracemem(a)
```

You are encouraged to read the help pages of the above R functions to learn more about their arguments, options and usage. **END** 

References

- [Moro et al., 2014] S. Moro, P. Cortez and P. Rita, A Data-Driven Approach to Predict the Success of Bank Telemarketing. Decision Support Systems, Elsevier, 62:22-31, June 2014
- Hadley Wickham. Memory. Advanced R, <http://adv-r.had.co.nz/memory.html>

By: Shakthi Kannan

The author is a free software enthusiast.